Version 1 explained

What the code in **version_1** does, the assumptions it makes, and the physics behind it

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Who this is for. Anyone on the team who needs to use, review or extend version 1 without reading every file. Sections 1–8 describe the code and how its parts connect. Appendix A derives every physical relation the code uses, Appendix B lists the symbols and Appendix C the sources. Where this report gives a number, it was produced by running the code (baseline configuration) or read from the source, not estimated.

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1. What version 1 does, in one page

Version 1 is a **conceptual-design optimizer** for a small, fast, four-propeller aircraft: a cigar-shaped body, four short pylon arms arranged as an X with one propeller on each tip, and optional canards near the nose. Given a mission (fly straight and level at **200 km/h** with a mass of **0.85 kg**) and a set of geometric, propulsive and stability limits, it finds the geometry, angle of attack and propeller speeds that give the **lowest aerodynamic drag** while the aircraft is:

- **in force balance**: lift (wings, body and propeller lift together) equals weight, and thrust along the flight path equals drag;
- **trimmed in pitch**: the total pitching moment about the CG is zero. Trim comes from *differential thrust* between the upper and lower propeller pairs, not from control surfaces;
- **statically stable**: the neutral point lies at least 2 % of the body length behind the CG;
- **buildable and within limits**: minimum thicknesses and clearances, motor power, propeller tip Mach number, the range of the propeller data, and local stall angles.

The aerodynamics come from AeroSandbox's **AeroBuildup** model (a fast, empirical, component-buildup method). The propellers come from **APC's published performance tables** for a 5×3.7 propeller. The whole model is written symbolically with CasADi, so the gradient-based solver **IPOPT** gets exact derivatives and converges in about 11 iterations (≈1 s).

Baseline result	Value	Baseline result	Value
Drag (= thrust)	1.308 N	Aerodynamic L/D	6.32
Fuselage length × diameter	286 × 73 mm	Angle of attack	3.25°
Pylon arm × chord (×4)	65 × 75 mm	Canard span × chord (×2)	114 × 60 mm
CG / neutral point from nose	50.0 / 55.7 mm	Propeller RPM upper / lower	29 835 / 30 219
Total shaft power	465 W	Propulsive efficiency	0.16

Two things to keep in mind when reading the results (both explained later): the 5×3.7 propeller is badly matched to 200 km/h (it runs close to windmilling, $\eta \approx 0.16$), and many dimensions are set by their limits rather than by aerodynamics (Section 5.4).

2. What is in version_1

Path	What it is	Used by a run?
`main.py`	The only place the pieces are assembled: assumptions → propeller data → optimizer → solve → report → figure.	entry point
`src/global_parameters/global_parameters.py`	**Parameters**: physical constants (g, ISA sea level). **Assumptions**: every design assumption and limit (switches, flight condition, structural, engine, mass, fuselage).	yes
`src/optimization/modules/`	The 7 phases of the model: propeller database, propulsion, design variables, aerodynamics, constraints, the optimizer that ties them, and the console report.	yes
`src/optimization/helpers/`	6 small functions: ellipse radius, polar→Cartesian, APC file reader, zero-thrust extension, thrust force split, thrust pitching moment.	yes
`data/PER3_5x37E-3_performance_file.txt`	APC performance tables (Ct, Cp vs J) for the 5×3.7 propeller, 1 000–40 000 RPM (repository root, shared by all versions).	yes
`data/5x37E-3-PERF_geometry.txt`	APC blade geometry (chord, pitch vs radius) for the same propeller.	no (reference)
`figures/`	Generated plots; a run writes `optimized_aircraft_three_view.png`.	output
`tests/optimization/`	70 pytests: helpers, modules, legacy behaviour, old-vs-new equivalence, golden reference data.	no
`src/stability/README.md`	Hand derivation of trim equations and an exact (no small-angle) neutral point for the cone-body concept. Theory notes; not code.	no
`src/deprecated/`	Earlier prototypes (Class I drag, quadcopter class, propeller drag) and the original single-file `optimizer.py`. Kept for reference only.	no
`tests/drag_estimation/`, `tests/stability/`	Placeholders for future subfolders (empty).	no

3. The vehicle and its coordinate systems

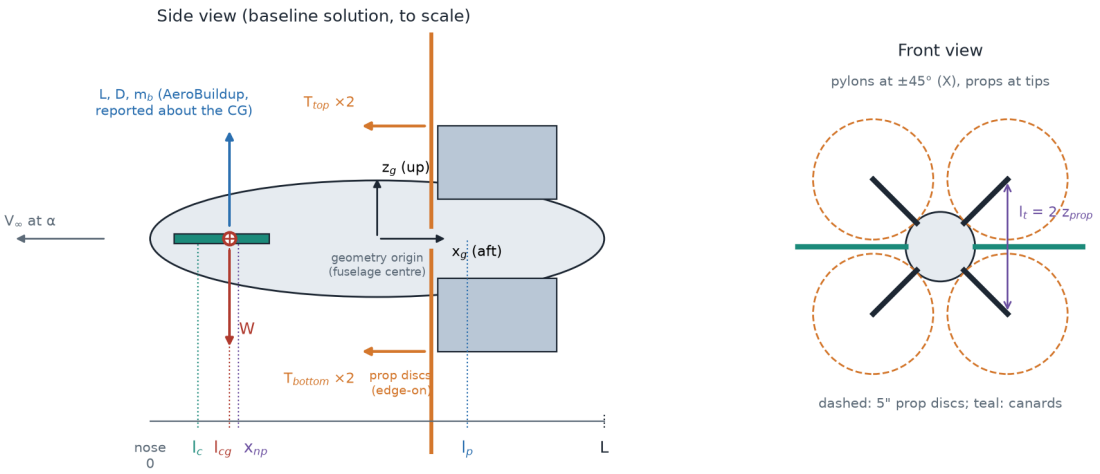


Figure 1. Baseline solution drawn to scale. Left: side view with the longitudinal stations; right: front view showing the X-shaped pylons, the propeller discs at the pylon tips and the canards.

Stations. The design variables `l_c`, `l_cg` and `l_p` are distances measured **aft from the nose**. `l_c` and `l_p` are quarter-chord (aerodynamic-centre) stations, so a surface's leading edge is at $l - c/4$ and its trailing edge at $l + 3c/4$.

Geometry axes (AeroSandbox's modelling frame) have their origin at the **fuselage centre**, +x pointing aft, +y to the right and +z up. A nose-based station l therefore sits at $x_g = l - L/2$; this conversion appears wherever geometry is built and when the neutral point is read back. **Body axes** point forward/right/down, and **wind axes** are body axes rotated by α so that x_w points into the oncoming flow. Lift and drag are defined in wind axes ($L = -F_{z,w}$, $D = -F_{x,w}$); pitching moments are reported in body axes, nose-up positive. Appendix A.2 gives the transformations.

Propellers. Each pylon is a constant-chord NACA 0012 arm whose root sits on the body surface at the pylon leading edge, at ±45° in the y–z plane. The four arms are generated as two mirrored half-wings (`pylons_upper`, `pylons_lower`). A propeller sits at each tip, so the upper pair is at $z = +z_{prop}$ and the lower pair at $-z_{prop}$, with $z_{prop} = (r_{root} + \text{pylon length})\cdot\sin 45°$. The two upper propellers share one RPM and the two lower ones another; making the lower pair push harder produces a nose-up moment, which is how the aircraft is trimmed.

4. How a run works

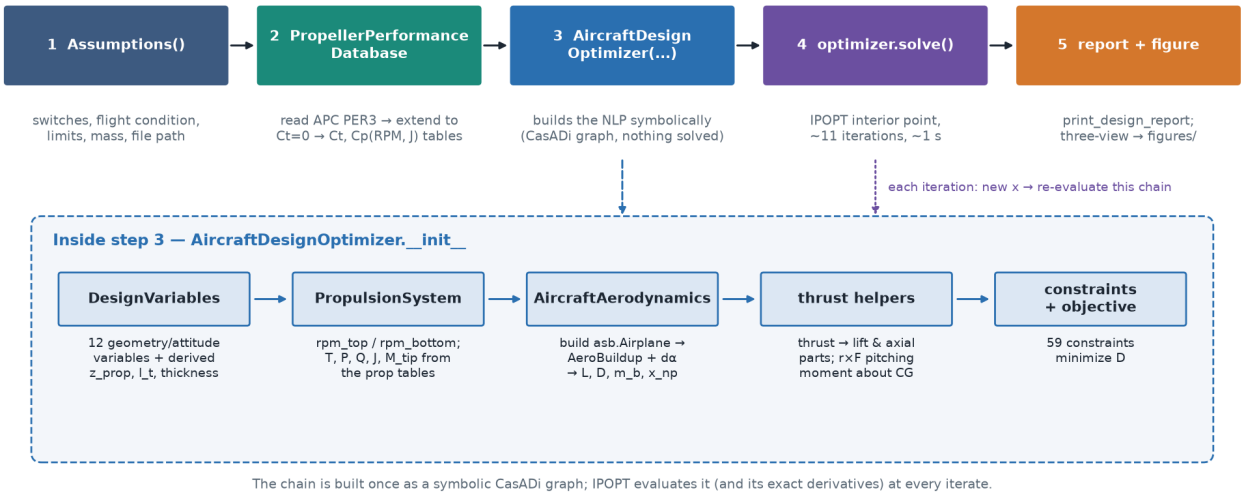


Figure 2. What happens when `python version_1/main.py` runs. Step 3 builds a symbolic model only; nothing is solved until step 4.

4.1 Step by step

Step 1 — Assumptions() (src/global_parameters/global_parameters.py)

Creates one object holding every assumption and limit (Section 6.2). Weight is a property, mass × `Parameters.G0`, so it always uses the shared constant.

Step 2 — PropellerPerformanceDatabase (modules/propeller_performance_database.py)

Reads the APC file with `read_apc_performance_blocks` (one block per RPM, 15 columns), keeps the blocks at or above `rpm_min` = 15 000 RPM (26 blocks, 15 000–40 000), extends each block in a straight line to zero thrust with `extend_to_zero_thrust`, and resamples all of them onto a shared grid of 89 advance ratios from 0 to $J_{max} = 0.881$. The result is two 26 × 89 tables, $C_T(\text{RPM}, J)$ and $C_P(\text{RPM}, J)$. Its `evaluate(rpm, V)` method turns them into thrust, shaft power, torque and J using a smooth 2-D B-spline (Appendix A.4–A.5).

Step 3 — AircraftDesignOptimizer(assumptions, prop_db) (modules/aircraft_design_optimizer.py)

Builds the optimization problem in this order:

- **DesignVariables** creates the 12 geometry/attitude variables with their bounds and initial guesses, and the derived quantities: pylon and canard thickness (chord × t/c), the pylon leading edge, the body radius at the pylon root (`radius_from_ellipse`), z_{prop} and the top-to-bottom propeller separation $l_t = 2z_{prop}$.
- **PropulsionSystem** creates `rpm_top` and `rpm_bottom` (one variable if differential thrust is off), the axial inflow $V\cos(\alpha + i_p)$, and calls the database for each pair: T, P, Q, J per propeller, pair totals, $\Delta T = 2T_{top} - 2T_{bottom}$, total power and tip Mach.
- **AircraftAerodynamics** builds an `asb.Airplane` (31-section ellipsoidal fuselage, the X pylons, the canards) with the moment reference at the CG, and runs AeroBuildup at (V, α) with the α-stability derivative. Output: L, D (profile + induced), the pitching moment m_b about the CG, $C_{L\alpha}$, $C_{m\alpha}$ and the neutral point x_{np}.

- **Thrust helpers**: `thrust_force_components` splits total thrust into a lift part and a part along the flight path; `thrust_pitching_moment` computes the pitching moment of the four thrust vectors about the CG (a differential part and an incidence part).
- **apply_design_constraints** adds the 31 design constraints (Section 5.3); the optimizer then sets the objective, minimize D.

Step 4 — solve()

Hands the problem to IPOPT through CasADi. Each iteration IPOPT proposes new variable values, the whole chain of step 3 is re-evaluated (with exact first and second derivatives), and the iteration stops when the optimality conditions are met to about 1e-8. If it fails, the violated constraints are printed.

Step 5 — report and figure

`print_design_report` prints drag, lift, geometry, stability, propulsion and incidence checks. `build_solved_airplane` rebuilds the aircraft from the numerical solution and `main.py` saves its three-view to `version_1/figures/`.

5. The optimization problem

5.1 Design variables

Symbol (code)	Meaning	Bounds	Initial	Optimum	At a limit?
α (AOA)	angle of attack	0 … 12°	4.00°	3.25°	
L (fuse_len)	fuselage length	200 … 600 mm	236 mm	286.2 mm	via tail margin
D_f (fuse_dia)	fuselage max diameter	50 … 150 mm	73.7 mm	73.4 mm	via diameter-at-station
l_{py} (pylon_len)	pylon arm length	50 … 200 mm	80 mm	65.0 mm	fixed = 65 mm
c_p (pylon_chord)	pylon chord	15 … 150 mm	75 mm	75.0 mm	via thickness ≥ 12 mm
i_p (pylon_inc)	pylon incidence = prop-axis tilt	−10 … 10°	−0.42°	-0.13°	
b_c (canard_len)	canard half-span (each)	30 … 300 mm	60 mm	114.0 mm	
c_c (canard_chord)	canard chord	60 … 200 mm	60 mm	60.0 mm	lower bound
i_c (canard_inc)	canard incidence	−6 … 10°	−2.00°	-0.60°	
l_c (l_c)	canard AC from nose	30 … 500 mm	30 mm	30.0 mm	lower bound
l_{cg} (l_cg)	CG from nose	50 … 500 mm	50 mm	50.0 mm	lower bound
l_p (l_p)	pylon AC from nose	80 … 600 mm	150 mm	200.0 mm	= l_{cg} + 150 mm
n_{top} (rpm_top)	upper-pair RPM	15 000 … 40 000	29 994	29835	
n_{bot} (rpm_bottom)	lower-pair RPM	15 000 … 40 000	30 287	30219	

The RPM upper bound is the smaller of `rpm_max` (44 000) and the highest RPM in the data file (40 000). Canard variables exist only when `use_canards` is on; `rpm_bottom` is the same variable as `rpm_top` when `use_diff_thrust` is off.

5.2 Objective

Minimize the total aerodynamic drag D from AeroBuildup (profile + induced). Propeller power is *not* in the objective; it only appears as the 1 kW-per-motor limit.

5.3 Constraints

There are 59 constraint rows in total: 28 come from the variable bounds above (AeroSandbox implements each bound as a scaled constraint) and 31 are design constraints. λ is the Lagrange multiplier at the baseline optimum; a non-zero λ means the constraint is **active** (it is pushing on the design). The last column is λ converted into drag: how much drag would be saved by relaxing the limit by one unit (Section 5.4).

Group	Constraint	Purpose	Baseline	Drag sensitivity
Geometry	$D_f \sqrt{(1 - ((l^* - L/2)/(L/2))^2)} \geq 70$ mm at $l^* = 100$ mm	body at least 70 mm wide 100 mm behind the nose	**active**	≈ 0.010 N per mm
	$l_{py} = 65$ mm (fix_pylon_len; else $\geq R + 10$ mm)	pylon length fixed / prop clears the body	**active**	
	$l_t = 2\, z_{prop} \geq D_{prop} + 10$ mm	upper and lower discs do not overlap	slack	—
	$c_p \cdot 0.16 \geq 12$ mm	pylon thick enough (structure, wiring)	**active**	≈ 0.029 N per mm
Force balance	$L_{aero} + L_{prop} = W$	steady level flight (≥ if steady_level off)	**active**	
	$T_{tot} \cos(\alpha + i_p) = D$	thrust along the flight path = drag	**active**	
Trim	$m_{b,aero} + M_{thrust} = 0$	zero pitching moment about the CG	**active**	
CG & stability	$l_{cg} = l_p - 150$ mm (fix_l_cg; else $l_{cg} \leq x_{np}$)	CG placed ahead of the pylons	**active**	
	$x_{np} - l_{cg} \geq 0.02\, L$	static margin ≥ 2 % of body length	**active**	≈ 0.0019 N per 0.1 % L
Layout	$l_p + 0.75\, c_p + 30$ mm $\leq L$	pylon trailing edge 30 mm ahead of the tail	**active**	
	$l_c + 0.75\, c_c \leq l_p - 0.25\, c_p$	canard ends before the pylon begins	slack	—
	$l_c + 20$ mm $\leq l_{cg}$	canard AC at least 20 mm ahead of the CG	**active**	
	$l_c - 0.25\, c_c \geq 0$	canard leading edge behind the nose	slack	—
Propulsion	$-0.5\, T_{tot} \leq \Delta T \leq 0.5\, T_{tot}$	differential-thrust authority	**active**	(lower side active)
	$\lvert\alpha + i_p\rvert \leq 12°$	propeller axis close to the flow	slack	—
	$0 \leq J \leq 0.881$ (both pairs)	stay inside the propeller data	slack	—
	$P \leq 1\,000$ W per motor	motor power	slack	—
	$M_{tip} \leq 0.80$	propeller tip Mach (noise, compressibility)	slack	—
	$T \geq 0$ (both pairs)	no windmilling	slack	—
Stall guards	$\lvert\alpha + i_p\rvert \leq 10°$, $\lvert\alpha + i_c\rvert \leq 10°$	geometric local incidence of pylons and canards	slack	—

5.4 What actually shapes the design

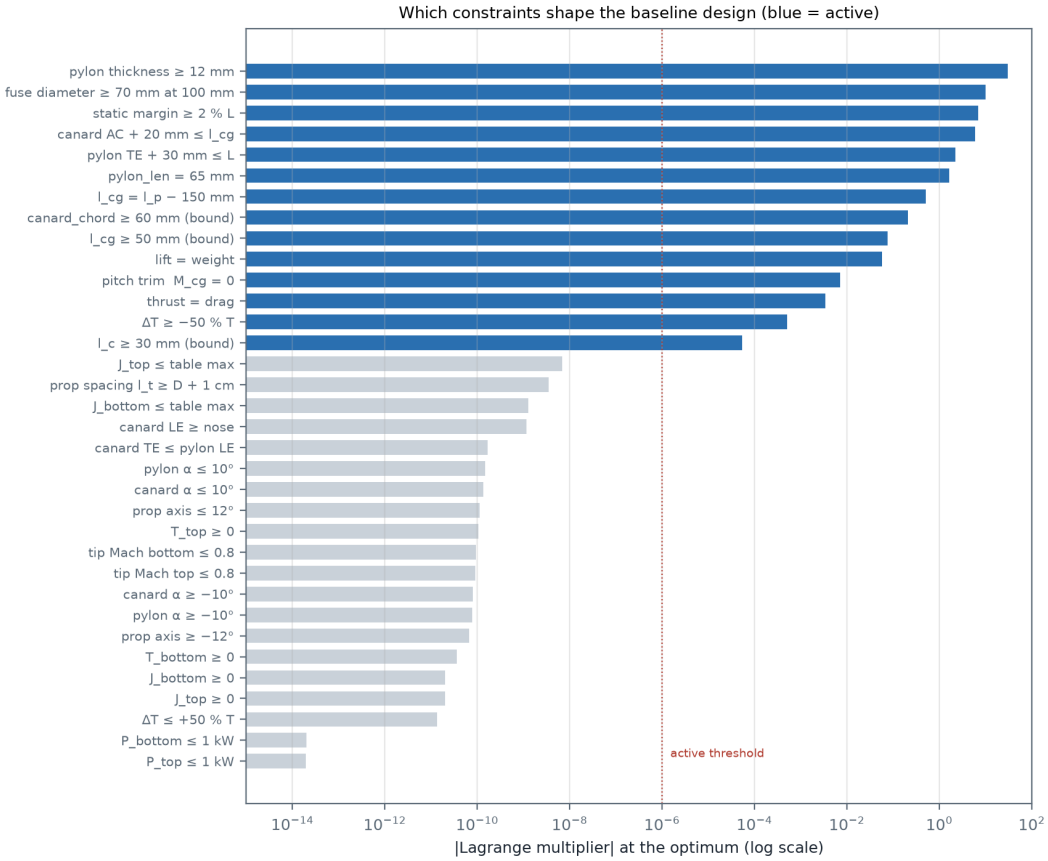


Figure 3. Absolute Lagrange multiplier of every constraint at the baseline optimum (log scale). Blue bars are active. A multiplier is the derivative of the optimal drag with respect to the constraint's limit, in newtons per unit of that constraint (for variable bounds, per unit of the variable divided by its initial guess, because AeroSandbox scales variables), so bars of different constraints are comparable only roughly.

A multiplier λ answers the question "*how much drag would I save if this limit were relaxed a little?*" (Appendix A.11). This was checked by re-solving: lowering the minimum pylon thickness from 12 to 11 mm reduced drag from 1.3083 to 1.2794 N (λ predicted 0.0299 N, actual 0.0289 N); lowering the minimum body diameter at 100 mm from 70 to 69 mm saved 0.0102 N (predicted 0.0099 N). Reading Figure 3 from the top:

- **Pylon thickness ≥ 12 mm** is the most expensive limit. It forces a 75 mm pylon chord, because the thickness is computed as chord × 0.16; every millimetre of required thickness costs about 0.03 N (2.2 % of drag).
- **Body diameter at 100 mm**, **static margin**, **canard–CG gap**, **tail margin** and the **fixed pylon length** together set the body: it is as short and slender as these allow.
- **CG at its 50 mm lower bound**, together with `fix_l_cg`, puts the pylons at exactly 200 mm. With no mass model, the optimizer moves the CG as far forward as allowed, because a forward CG gives stability for free.
- **Differential thrust is at its −50 % limit**: the lower pair pushes three times as hard as the upper pair. More trim authority would let the canard work less.
- Motor power, tip Mach, the propeller data range, stall and propeller-axis limits all have plenty of slack.

6. Assumptions

Assumptions come in three kinds: the **physics and modelling** choices built into the method (6.1), the **numerical values** stored in `Assumptions` (6.2) and the **physical constants** in `Parameters` (6.3). Section 6.4 lists places where they are not fully consistent with each other.

6.1 Modelling assumptions

M1 **Steady, straight, level flight** at one design point (200 km/h, sea level); no climb, turn or gust cases. (`steady_level=False` turns the balances into inequalities, i.e. allows climb or acceleration.)

M2 **Longitudinal plane only.** Side force, roll and yaw are not modelled: the aircraft is assumed left–right symmetric with left and right propellers of a pair at the same RPM, so their lateral forces and moments cancel.

M3 **Propeller torque reactions and gyroscopic effects cancel** (counter-rotating pairs are implied); propeller normal force ("P-factor"), slipstream and propwash on the pylons and body are ignored.

M4 **Each propeller only sees the axial inflow** $V\cos(\alpha + i_p)$; the in-plane component of the flow is neglected. Its performance comes from isolated-propeller wind-tunnel-style data (APC), with no installation effects.

M5 **APC data are trusted** and are extended linearly over a short gap (≤ 0.02 in J) to zero thrust; between data points they are interpolated with a smooth B-spline.

M6 **Thrust acts at the pylon quarter-chord station** l_p and at $z = \pm z_{prop}$, along the pylon chord line (tilted by the pylon incidence i_p).

M7 **AeroBuildup aerodynamics**: each component (body, pylons, canards) is analysed on its own and the results are summed. No canard downwash on the pylons, no body–wing interference beyond root placement, no ground effect.

M8 **All lifting surfaces use the NACA 0012 section**, constant chord, no twist, no sweep.

M9 **The body is a closed prolate ellipsoid** (31 sections), so it has no base area and no potential-flow lift; its lift is viscous crossflow and its moment is the destabilizing Munk moment (Appendix A.8).

M10 **Static stability is judged by the aerodynamic neutral point only** ($\partial C_m/\partial\alpha$ from AeroBuildup), evaluated at the trim α with the propellers' contribution excluded, stick-fixed (ΔT held constant).

M11 **Stall is guarded geometrically**: local incidence = α + surface incidence, limited to ±10°. NeuralFoil itself also models section stall, so this is an additional safety margin.

M12 **No mass model.** The CG station is a free design variable (or tied to the pylons), and the mass is a fixed 0.85 kg regardless of geometry.

M13 **Sea-level standard atmosphere**, no wind.

M14 **Drag is the objective**; power, energy and efficiency are outputs, not goals.

6.2 Values in Assumptions

Field	Value	Meaning	Used in
fix_l_cg / fix_pylon_len / fix_fuse_dia	True / True / True	tie CG to pylons; fix pylon length; enforce diameter at a station	constraints
use_canards / use_diff_thrust / steady_level	True / True / True	canards present; separate RPM per pair; equality balances	several
velocity	55.56 m/s (200 km/h)	design airspeed	optimizer, report
static_margin_frac	0.02	required $(x_{np} - x_{cg})/L$	constraints
alpha_stall_deg / prop_axis_max_deg	10° / 12°	local incidence and prop-axis limits	constraints
tc_pylon / tc_canard	0.16 / 0.12	thickness-to-chord for the thickness calculation	design variables
pylon_thick_min	12 mm	minimum pylon thickness	constraints
perf_file	data/PER3_5x37E-3 …	APC performance file	main
prop_dia (prop_radius)	127 mm (63.5 mm)	propeller diameter (5 in)	main, propulsion, constraints
rpm_min / rpm_max	15 000 / 44 000	RPM range (upper capped at 40 000 by the data)	main, propulsion
P_motor_max / M_tip_max	1 000 W / 0.80	per-motor shaft power; tip Mach	constraints
max_diff_frac	0.5	$\lvert\Delta T\rvert \le 0.5\ T_{tot}$	constraints
prop_spacing_gap / prop_clearance_gap	10 mm / 10 mm	gap between upper/lower discs; prop tip to body	constraints
aircraft_mass (aircraft_weight)	0.85 kg (8.336 N)	mass; weight = mass × G0	constraints, report
cg_offset	150 mm	$l_{cg} = l_p - \text{cg_offset}$	design variables
fuse_dia_min / l_min_dia	70 mm / 100 mm	minimum body diameter and its station	constraints
forced_pylon_len	65 mm	fixed pylon arm length	constraints
tail_margin / canard_cg_gap	30 mm / 20 mm	pylon TE to tail; canard AC to CG	constraints
n_propellers	4	number of propellers	not used

Variable bounds and initial guesses are solver settings rather than physical assumptions, so they stay next to the variables in `DesignVariables` and `PropulsionSystem` (Table 5.1).

6.3 Physical constants (Parameters)

Constant	Value	Used for
G0	9.80665 m/s²	weight
AIR_DENSITY_SEA_LEVEL	1.225 kg/m³	propeller thrust and power
GAMMA_AIR, GAS_CONSTANT_AIR, TEMPERATURE_SEA_LEVEL	1.4, 287.05 J/(kg K), 288.15 K	speed of sound $a = \sqrt{\gamma RT}$ = 340.29 m/s (tip Mach)
PRESSURE_SEA_LEVEL, DYNAMIC_VISCOSITY_SEA_LEVEL	101 325 Pa, 1.789e-5 kg/(m s)	not used by the optimizer

6.4 Known inconsistencies (reported, not changed)

- **Pylon thickness.** The thickness constraint assumes t/c = 0.16, but the aerodynamics use NACA 0012 (t/c = 0.12) on every surface. Either the real pylon is thicker than the aerodynamics assume (so drag is underestimated), or the structural constraint is stricter than the real part. Because this is the most active constraint, it matters.
- **Two atmospheres.** The propellers and tip Mach use `Parameters` (ρ = 1.225, a = 340.29 m/s); `asb.OperatingPoint` uses AeroSandbox's own ISA model (ρ = 1.2249992, a = 340.2941 m/s). The difference is below 1e-5 and has no visible effect, but the values come from two places.
- **Unused values.** `n_propellers` is never read ("two per pair" is hard-coded), and the canard thickness is computed but not constrained.

- **Reference area and chord.** AeroSandbox takes S_{ref} and c_{ref} from the first wing (`pylons_upper`), so the reported $C_{L\alpha}$ and $C_{m\alpha}$ are normalized by one pylon pair. Forces, moments and x_{np} do not depend on this choice.

7. Where everything is used

7.1 Dependency graph

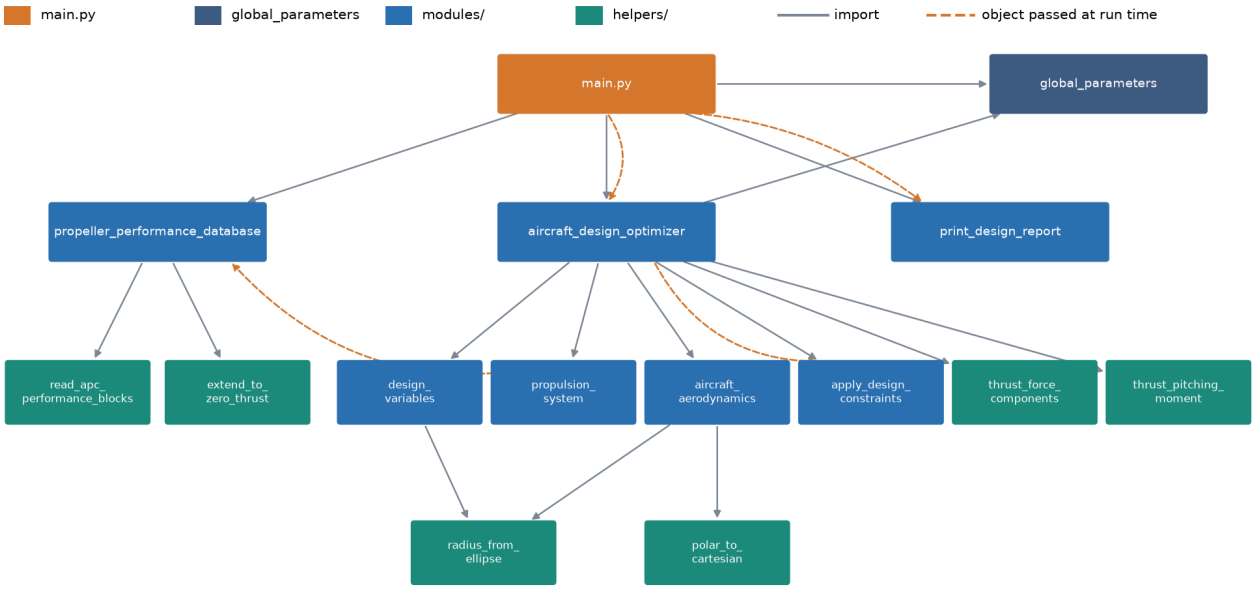
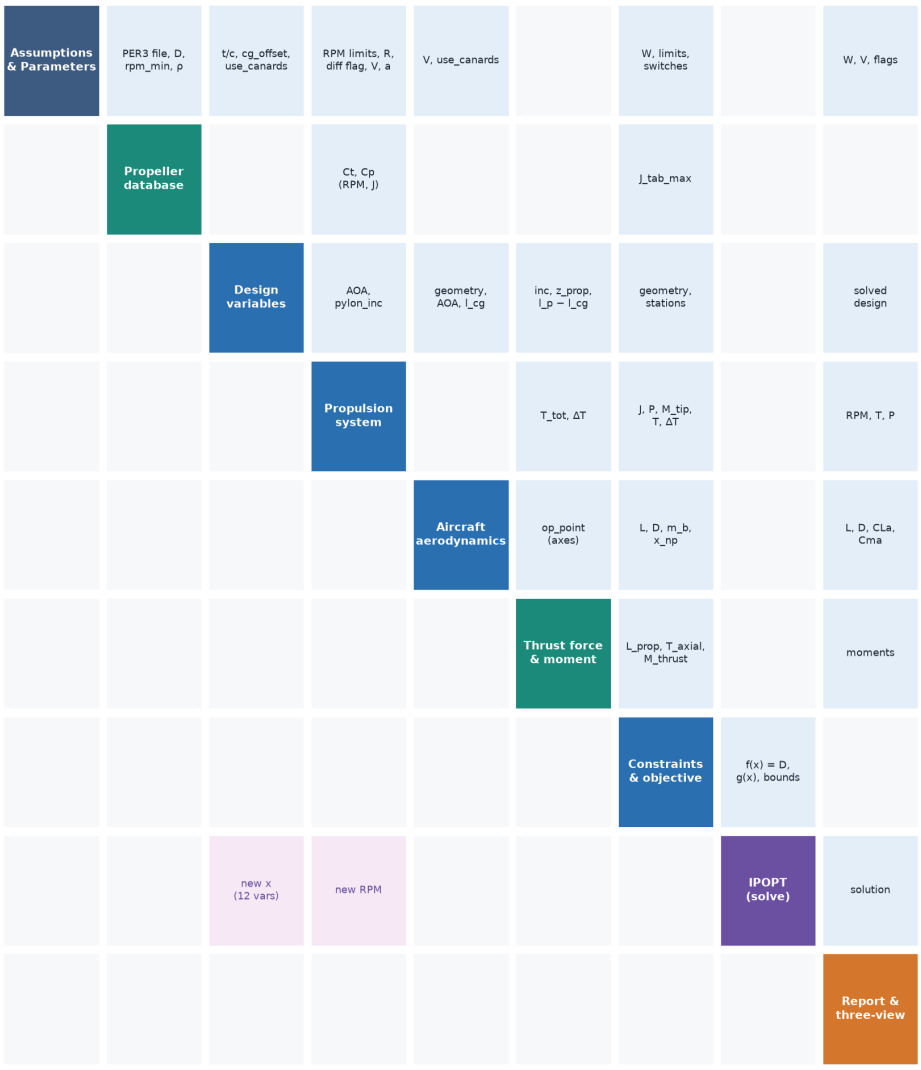


Figure 4. Who imports whom (grey, parsed from the source code) and which objects are handed over while the program runs (orange dashed). Arrows point from the user to the thing used.

`main.py` is the only file that wires the parts together. The optimizer imports the four phase modules and the two thrust helpers; the geometry helpers are shared by `design_variables` and `aircraft_aerodynamics`. The propeller database is created in `main.py` and handed to the optimizer, which passes it on to the propulsion system; `apply_design_constraints` receives the whole optimizer object. No module imports `main.py` and there are no import cycles.

7.2 N2 chart (data flow)



Read row i → column j: output of component i used as input by component j. Upper triangle (blue) = feed-forward; lower triangle (purple) = feedback (IPOPT iterate).

Figure 5. N2 chart. The diagonal holds the components in execution order. A cell in row i, column j names what component i passes to component j. Above the diagonal: feed-forward; below: feedback from the solver.

Reading the chart: the **Assumptions** row shows that settings feed almost every component. The **Design variables** row feeds propulsion, aerodynamics, the thrust helpers and the constraints, which is why geometry changes ripple through everything. The only feedback is from **IPOPT** back to the design variables and RPMs: the model itself is a single forward pass, with no internal iteration loops, which is what makes it fast and exactly differentiable.

7.3 Which file uses which assumption

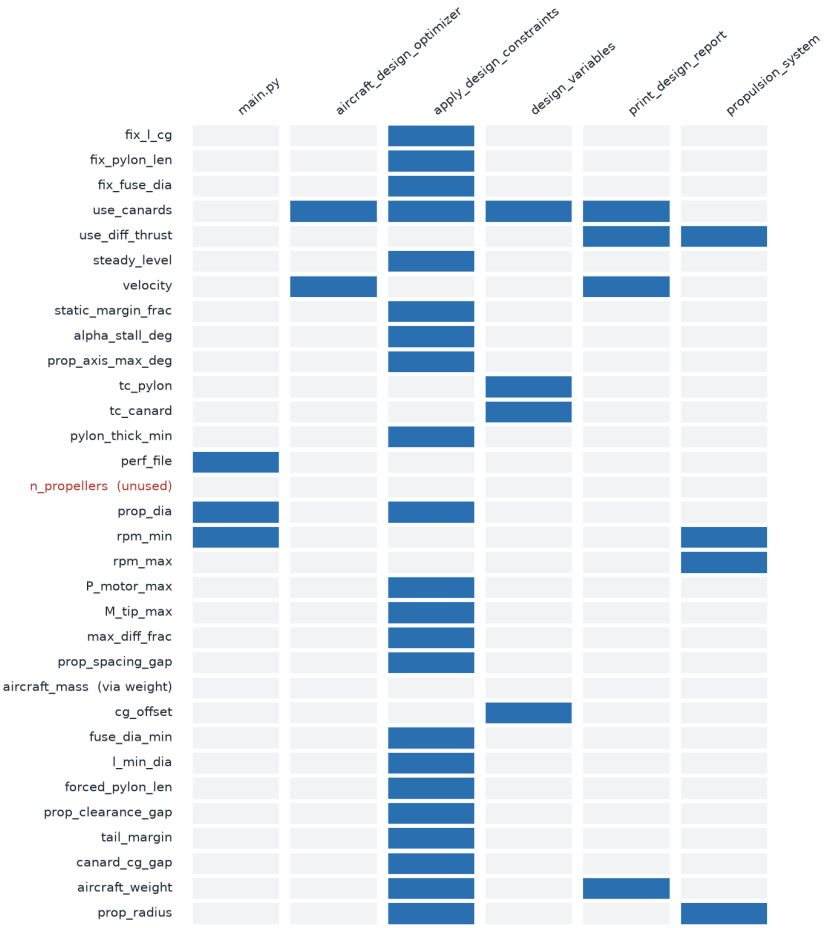


Figure 6. Each Assumptions field (row) against the files that read it (columns), found by scanning the source. Files that read no assumptions directly are not shown.

8. Result, tests and how to run

Is the result physical? Lift equals weight (8.336 N), thrust along the flight path equals drag (1.308 N), and the pitching moment is about 1e-15 N·m. The aerodynamic pitching moment about the CG is nose-down (−0.047 N·m), and it is cancelled by the lower propeller pair pushing harder (ΔT = −0.655 N). An L/D around 6 is plausible for a body-lift vehicle with small surfaces. The weak point is the propeller: at J ≈ 0.87–0.88 it is close to its zero-thrust point, so 465 W of shaft power produce only 73 W of useful power (η ≈ 0.16, see Figure 8).

Tests. `version_1/tests/optimization/` holds 70 tests (all passing). They cover every helper, every module, the physics invariants of the solved design (force and moment balance, all constraints, a hand-computed r × F moment), the original single-file code, and a golden reference of five configurations.

```
source .venv/bin/activate
python version_1/main.py                       # optimize, print the report, save the three-view
python -m pytest version_1/tests/optimization  # 70 tests, about 90 s
```

To study another case, change a field of `Assumptions` before building the optimizer, for example `asm = Assumptions(); asm.use_canards = False; run_design_optimization(asm)`.

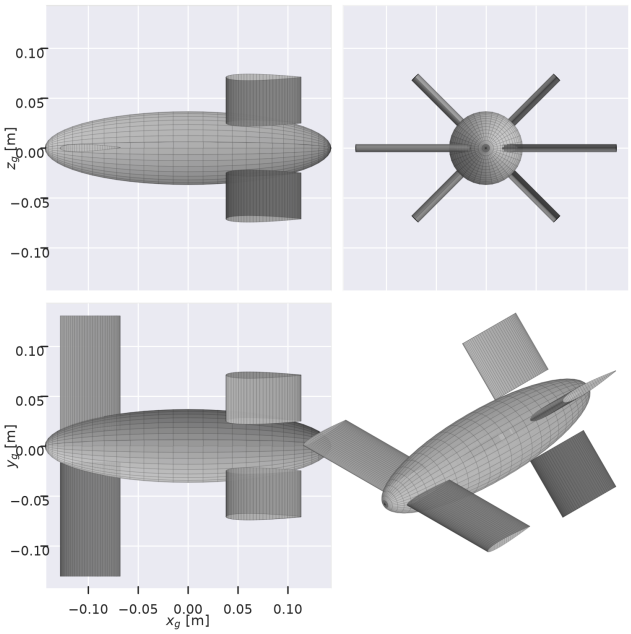


Figure 7. Three-view of the baseline optimum (`version_1/figures/optimized_aircraft_three_view.png`).

Appendix A. Theory

This appendix derives the relations used by the code, in the order data flows through a run. Symbols are listed in Appendix B.

A.1 Atmosphere and flow parameters

The International Standard Atmosphere at sea level gives T = 288.15 K, p = 101 325 Pa and ρ = 1.225 kg/m³. The speed of sound follows from the ideal-gas relation:

$$a = \sqrt{(\gamma R T)} = \sqrt{(1.4 \cdot 287.05 \cdot 288.15)} = 340.29 \text{ m/s}$$

At V = 55.56 m/s this gives:

$$M = V/a = 0.163,\ q = \tfrac{1}{2} \rho V^2 = 1\,890 \text{ Pa},\ Re_{\ell} = \rho V \ell / \mu$$

so Re ≈ 1.1·10⁶ on the body length, 2.9·10⁵ on the pylon chord and 2.3·10⁵ on the canard chord. These are low Reynolds numbers where laminar-separation effects matter, which is why AeroBuildup uses NeuralFoil (A.7) rather than a fixed lift slope. M < 0.3, so compressibility is minor (it is still included by AeroBuildup).

A.2 Coordinate systems and transformations

Three right-handed frames are used. Geometry axes (g): origin at the fuselage centre, x aft, y right, z up. Body axes (b): x forward, y right, z down. The two differ by a 180° rotation about y:

$$x_b = -x_g,\ y_b = y_g,\ z_b = -z_g$$

Wind axes (w) are body axes rotated by the angle of attack α about y (sideslip β = 0 here):

$$x_w = x_b \cos\alpha + z_b \sin\alpha,\ y_w = y_b,\ z_w = -x_b \sin\alpha + z_b \cos\alpha$$

so x_w points into the oncoming flow. Lift and drag are defined as $L = -F_{z,w}$ and $D = -F_{x,w}$. This was verified numerically: at α = 10°, a unit forward force in geometry axes becomes (0.985, 0, −0.174) in wind axes, i.e. +0.985 along the flight path and +0.174 of lift. A pitching moment about y is unchanged between geometry and body axes (the y axes coincide), and is positive nose-up.

A.3 Geometry

Fuselage

The body is a prolate ellipsoid with semi-axes a = L/2 and b = D_f/2. At axial position x (from the centre) its radius is

$$r(x) = b \sqrt{(1 - (x/a)^2)}$$

The 31 cross-sections use cosine spacing, $x_i = -a \cos\beta_i$ with $\beta_i = i\pi/30$, which clusters sections near the nose and tail where the curvature is highest. Substituting gives $r_i = b \sin\beta_i$ exactly. The code uses this form because the derivative of $\sqrt{(1 - (x/a)^2)}$ is infinite at the tips (x = ±a), which made the solver fail before the fix; $b \sin\beta_i$ has a finite derivative everywhere.

Pylons and propellers

The pylon leading edge is at $l_p - c_p/4$ (nose-based), i.e. $x_{LE} = l_p - c_p/4 - L/2$. The root lies on the body surface at $r_{root} = r(x_{LE})$ and the tip at $r_{root} + l_{py}$, both along the ±45° directions. The propeller hubs therefore sit at

$$z_{prop} = (r_{root} + l_{py}) \sin 45°,\ l_t = 2\, z_{prop}$$

Upper and lower discs on the same side are l_t apart, as are left and right discs of a pair, so $l_t \geq D_{prop}$ + gap prevents any disc overlap.

Diameter-at-station constraint

To guarantee a diameter of at least d_{min} at the station l* from the nose, the ellipse must satisfy $D_f \sqrt{(1 - ((l^* - a)/a)^2)} \geq d_{min}$, which the code writes as

$$D_f \geq d_{min} / \sqrt{(1 - ((l^* - L/2)/(L/2))^2)}$$

A.4 Propeller theory

Dimensional analysis of a propeller with diameter D turning at n revolutions per second in a flow of speed V, density ρ, viscosity μ and speed of sound a shows that thrust and power can be written in non-dimensional form:

$$C_T = T / (\rho\, n^2\, D^4),\ C_P = P / (\rho\, n^3\, D^5),\ \text{with } C_T, C_P = f(J, Re, M_{tip})$$

$$J = V / (n\, D)$$

The advance ratio J is the distance the propeller advances per revolution in diameters. As J grows, the blade sections see a smaller angle of attack, so C_T falls and reaches zero at the zero-thrust advance ratio J_0 (≈ 0.88 for this propeller). APC tabulates C_T and C_P for many RPMs; tabulating against RPM captures the Reynolds and tip-Mach dependence. From the coefficients:

$$T = C_T\, \rho\, n^2\, D^4,\ P = C_P\, \rho\, n^3\, D^5,\ Q = P / (2\pi\, n),\ \eta = T\, V / P = J\, C_T / C_P$$

Only the flow component along the propeller axis is used, $V_{ax} = V \cos(\alpha + i_p)$, because the propeller axis is tilted by $\alpha + i_p$ from the flow. The blade tip sees the rotational and axial speeds combined:

$$M_{tip} = \sqrt{((2\pi\, n\, R)^2 + V_{ax}^2)} / a$$

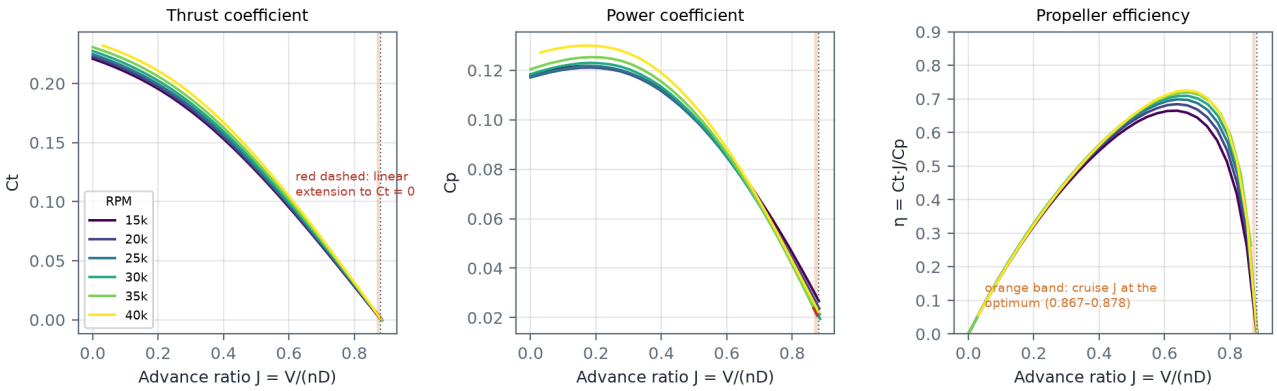


Figure 8. APC data for the 5×3.7 propeller at six of the 26 RPMs used. The red dashed segments are the linear extensions to $C_T = 0$; the orange band is where the optimum operates, just short of zero thrust, where efficiency falls steeply.

A.5 Propeller data processing

For each RPM block, a straight line is fitted by least squares through the last three (J, C_T) points, $C_T \approx mJ + c$, giving $J_0 = -c/m$. If J_0 is beyond the last data point and the last C_T is still positive, the point (J_0, 0, $C_P(J_0)$) is appended, with C_P extrapolated the same way. All blocks are then resampled linearly onto a common grid $J \in [0, \min J_0] = [0, 0.881]$ of 89 points. Without the extension, the shortest block (35 000 RPM, ending at J = 0.864) capped the grid at 0.86; at 200 km/h that forbade thrust below about 2.8 N and forced the optimizer to raise drag to match.

Inside the optimizer, C_T and C_P are evaluated with a 2-D cubic B-spline over (RPM, J). A B-spline is twice continuously differentiable, which a gradient-based solver needs (A.11); linear interpolation would have kinks at every grid line. The constraints $0 \le J \le 0.881$ and the RPM bounds keep every evaluation inside the table.

A.6 Thrust forces and pitching moment

Each propeller's thrust acts along its axis, which is tilted up by the pylon incidence i_p relative to the body. In geometry axes (x aft, z up), a propeller producing T gives

$$F = T \cdot (-\cos i_p,\ 0,\ \sin i_p)$$

Summing the four propellers, $T_{tot} = 2T_{top} + 2T_{bottom}$, and rotating into wind axes (A.2), the thrust axis makes the angle $\alpha + i_p$ with the flow, so

$$L_{prop} = T_{tot} \sin(\alpha + i_p),\ T_{axial} = T_{tot} \cos(\alpha + i_p)$$

(the code performs the rotation with AeroSandbox's axis conversion rather than this closed form; a test checks that the two agree for nine combinations of α and i_p). For the pitching moment, place the

propellers at $\Delta x = l_p - l_{cg}$ aft of the CG and at $z = \pm z_{prop}$. The y-component of $r \times F$ is $M_y = z\,F_x - x\,F_z$. Summing the upper pair ($z = +z_{prop}$, thrust $2T_{top}$) and the lower pair ($z = -z_{prop}$, thrust $2T_{bottom}$):

$$M_y = z_{prop}(-2T_{top}\cos i_p) - z_{prop}(-2T_{bottom}\cos i_p) - \Delta x\,(2T_{top} + 2T_{bottom})\sin i_p$$

$$\mathbf{M_{thrust} = -z_{prop}\,\Delta T\cos i_p - \Delta x\,T_{tot}\sin i_p},\ \Delta T = 2T_{top} - 2T_{bottom}$$

The first term is the **differential part**: if the lower pair pushes harder ($\Delta T < 0$), the moment is positive, nose-up. The second is the **incidence part**: an upward-tilted thrust acting behind the CG pushes the tail up, nose-down. The left–right offsets cancel by symmetry. At the baseline the differential part is +0.0465 N·m and the incidence part +0.0004 N·m.

A.7 AeroBuildup: lifting surfaces

AeroBuildup is a component-buildup method: it analyses each wing and body separately and sums forces and moments about the reference point (here the CG). For a lifting surface:

- **Section aerodynamics.** Each spanwise section's lift, drag and moment coefficients (c_l, c_d, c_m) come from **NeuralFoil**, a neural-network surrogate trained on XFoil viscous panel solutions, at the section's own Reynolds number, angle of attack and Mach number. This includes viscous drag and stall.
- **Finite-span lift.** The 2-D lift is reduced for finite aspect ratio A with the DATCOM/Helmbold relation (Raymer, Sect. 12.4.1):

$$C_L/c_l = A\,/\,(2 + \sqrt{(4 + A^2\beta^2/\eta^2 + (A\tan\Lambda/\eta)^2)}),\ \beta^2 = 1 - M^2,\ \eta = 1$$

which tends to the lifting-line value $A/(A+2)$ for large A and to slender-wing theory for small A.

- **Effective span for a dihedral surface.** A symmetric surface with dihedral Γ gets an effective span blended between a flat wing and a vertical fin, $b_{eff} = b_0 + (b_{90} - b_0)\sin^2\Gamma$ (similarly for area). The X pylons ($\Gamma = \pm 45°$) sit exactly half-way: they act partly as a wing and partly as fins.
- **Induced drag.** Computed once for the whole aircraft from the total lift L and side force Y:

$$D_i = (L^2 + Y^2)\,/\,(q\,\pi\,b_e^{\,2}),\ b_e^{\,2} = \text{softmax over components of } (b_{eff}^{\,2}\,e)$$

where e is the Oswald efficiency (Nita–Scholz model for wings, 0.95 for the body). The smooth maximum means the component with the largest effective span dominates. Profile drag is the integral of section c_d. Not modelled: canard downwash on the pylons, propwash, and wing–body interference beyond placing each root on the body surface.

A.8 AeroBuildup: fuselage

The body is treated with slender-body theory plus a viscous crossflow correction (Jorgensen, NASA TR R-474):

- **Potential flow (Munk, Drela FVA Eq. 6.77–6.78).** For a slender body the inviscid normal force is $N = \rho V^2\,S_{base}\sin\alpha\cos\alpha$, which is zero for a closed body like this ellipsoid ($S_{base} = 0$). What remains is a pure couple, the **Munk moment**, $M \approx \rho V^2\,\mathrm{Vol}\sin\alpha\cos\alpha$ (Vol = body volume), which is nose-up and therefore destabilizing. This is the main reason the neutral point lies only a few millimetres behind the CG.
- **Viscous crossflow.** Each slice of the body is treated as a cylinder in the crossflow $V\sin\alpha$, giving a normal force per unit length $\tfrac{1}{2}\rho\,(V\sin\alpha)^2 \cdot 2r \cdot c_{d,n}$, with $c_{d,n}$ the drag coefficient of a circular cylinder at the crossflow Reynolds and Mach numbers. This is the body's lift, and it grows as $\sin^2\alpha$.
- **Profile drag.** $D = q\,(C_f \cdot FF \cdot S_{wet} + C_{D,base}\,S_{base} + \text{wave drag})$, with $C_f = (3.46\log_{10}\mathrm{Re} - 5.6)^{-2}$ and a form factor FF that depends on the fineness ratio L/D_f. At M = 0.16 the wave-drag term is negligible.

A.9 Trim equations

With forces in wind axes and moments about the CG, steady level flight requires

$$L_{aero}(\alpha, x) + T_{tot}\sin(\alpha + i_p) = W$$

$$T_{tot}\cos(\alpha + i_p) = D(\alpha, x)$$

$$m_{b,aero}(\alpha, x) - z_{prop}\,\Delta T\cos i_p - \Delta x\,T_{tot}\sin i_p = 0$$

These play the role of the trim equations in Section 3 of `src/stability/README.md`, with two differences. First, thrust is resolved exactly with the sine and cosine of the thrust-axis angle $\alpha + i_p$ (the README's force equations use $\cos\alpha$ in both directions). Second, AeroBuildup supplies L, D and the moment of all aerodynamic forces about the CG, including each component's own moment about its aerodynamic centre, so the README's separate L_s, L_p and M_{ac} terms are already summed inside $m_{b,aero}$, and the drag moment arms that the README drops are kept.

A.10 Static stability and the neutral point

The aircraft is statically stable in pitch if a nose-up disturbance in α creates a nose-down restoring moment, $dC_m/d\alpha < 0$ about the CG. The neutral point is the CG position at which $dC_m/d\alpha = 0$. Because moving the reference point by Δx changes C_m by $C_L \cdot \Delta x / c_{ref}$, it follows that

$$x_{np} = x_{ref} - (C_{m\alpha} / C_{L\alpha})\, c_{ref}$$

AeroBuildup computes $C_{L\alpha}$ and $C_{m\alpha}$ by finite differences (α perturbed by 0.001°) and returns x_{np} in geometry axes; the code adds L/2 to make it nose-based. c_{ref} and S_{ref} cancel in this ratio, so the result does not depend on the reference values. This is the same definition as the README's Section 5 (the x where $\partial M/\partial\alpha = 0$), evaluated by AeroBuildup at the trim α with every trigonometric term included. The static margin is $SM = x_{np} - x_{cg}$, required to be at least 0.02 L.

At the baseline, $C_{m\alpha} = -0.690$ /rad and $C_{L\alpha} = 9.04$ /rad (both based on one pylon pair's area and chord), $x_{np} = 55.7$ mm and $x_{cg} = 50.0$ mm, so SM = 5.7 mm = 2.0 % L, exactly the minimum.

A.11 Numerical optimization

The design problem is a nonlinear program (NLP):

$$\text{minimize } f(x) = D(x) \text{ subject to } g_L \le g(x) \le g_U,\ x_L \le x \le x_U$$

with 14 variables and 59 constraint rows. **IPOPT** solves it with a primal-dual interior-point method: inequality constraints get slack variables $s \ge 0$, which are kept positive by a logarithmic barrier,

$$\text{minimize } f(x) - \mu \Sigma \ln s_i$$

Newton's method is applied to the optimality (KKT) conditions of this barrier problem while μ is driven towards zero. At the solution, the KKT conditions hold:

$$\nabla f(x^*) + \Sigma \lambda_i \nabla g_i(x^*) = 0,\ \lambda_i \cdot (\text{slack of constraint } i) = 0$$

The second condition (complementarity) says a multiplier is non-zero only for active constraints. The multiplier is also the sensitivity of the optimum to the limit, $\lambda_i = -\partial f^*/\partial(\text{limit}_i)$, which is how Section 5.4 converts multipliers into newtons of drag; the re-solves quoted there confirm it to within 4 %.

Newton's method needs first and second derivatives. **CasADi** provides them exactly by automatic differentiation of the whole model graph, including AeroBuildup and the NeuralFoil network. This places one strict requirement on the model: every expression must be smooth and finite over the region the solver explores. That is why the code uses a B-spline instead of linear interpolation, why AeroSandbox uses a smooth softmax instead of max(), and why the $\sqrt{0}$ in the old fuselage radius (infinite derivative, hence NaN) stopped the solver at iteration 0. AeroSandbox scales each variable by the size of its initial guess, so variables of very different size (RPM $\approx 3 \cdot 10^4$, lengths $\approx 10^{-1}$ m) are balanced. IPOPT finds a local optimum; the five reference configurations and the robustness runs all converged from the same initial guess.

Appendix B. Nomenclature

Symbol	Meaning	Unit	Symbol	Meaning	Unit
a	speed of sound	m/s	L	fuselage length; lift (context)	m; N
A	aspect ratio	–	l_c, l_{cg}, l_p	canard AC, CG, pylon AC from nose	m
b, b_{eff}	span, effective span	m	l_t	upper-to-lower prop separation	m
c_p, c_c	pylon / canard chord	m	M, M_{tip}	Mach number, tip Mach	–
C_T, C_P	thrust / power coefficient	–	m_b	aero pitching moment about CG	N·m
$C_{L\alpha}$, $C_{m\alpha}$	lift / moment slope	1/rad	n	propeller speed	rev/s
D	drag; prop diameter (context)	N; m	P, Q	shaft power, torque	W, N·m
D_f	fuselage diameter	m	q	dynamic pressure ½ρV²	Pa
e	Oswald efficiency	–	R	prop radius; gas constant	m; J/(kg K)
i_p, i_c	pylon / canard incidence	rad	T, T_{tot}, ΔT	thrust per prop, total, upper−lower pair	N
J, J_0	advance ratio, zero-thrust J	–	V, V_{ax}	airspeed, axial inflow	m/s
W	weight	N	x_{np}	neutral point from nose	m
z_{prop}	prop height above axis	m	α	angle of attack	rad
γ	ratio of specific heats	–	η	propeller efficiency	–
λ	Lagrange multiplier	N per unit	ρ, μ	density, viscosity	kg/m³, kg/(m s)

Appendix C. References

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